

# POLYP DETECTION MEDICAL REPORT

Analysis Date: June 04, 2026

## VIDEO INFORMATION

|               |                                      |
|---------------|--------------------------------------|
| Video ID:     | 8af3fb2b-7916-4bbd-bd76-c1dd303444d0 |
| Total Frames: | 3006                                 |
| Frame Rate:   | 25.0 fps (assumed)                   |
| Duration:     | 120.2 seconds                        |

## DETECTION SUMMARY

| Detection Method              | Count        | Status            |
|-------------------------------|--------------|-------------------|
| YOLO Detections               | 984          | ✓                 |
| RT-DETR Detections            | 1961         | ✓                 |
| MedSAM2 Detections            | 2065         | ✓                 |
| Consensus (3-Model Agreement) | 4            | ✓                 |
|                               |              |                   |
| <b>Risk Classification</b>    | <b>Count</b> | <b>Percentage</b> |
| HIGH RISK                     | 0            | 0.0%              |
| MEDIUM RISK                   | 0            | 0.0%              |
| LOW RISK                      | 4            | 100.0%            |
| TOTAL ANALYZED                | 4            | 100.0%            |

DETAILED POLYP ANALYSIS

POLYP #1 — LIFTED\_POLYP [LOW\_RISK] (Tracked across 164 frames)



| Feature           | Value                                    | Interpretation                                           |
|-------------------|------------------------------------------|----------------------------------------------------------|
| Frame Index       | 1075                                     | Frame 1075 of 3006                                       |
| Detection Method  | Detector Consensus (MedSAM2 Unavailable) | MedSAM2 Unavailable (MedSAM2 unavailable for this video) |
| POLYP TYPE        | LIFTED_POLYP                             | Type confidence: 60.3%                                   |
| Redness Score     | 0.021                                    | Color-based vascularization                              |
| Relative Radius   | 9.7%                                     | Polyp size as % of frame diagonal                        |
| Texture Score     | 0.093                                    | Surface morphology                                       |
| Vessel Visibility | 0.000                                    | Vascularization indicator                                |
| Risk Tier         | LOW RISK                                 | Validated by ESGE 2017 / NICE classification             |
|                   |                                          |                                                          |

|                         |                   |                                                         |
|-------------------------|-------------------|---------------------------------------------------------|
| Clinical Classification | HIGH_RISK         | Final symbolic reasoning bucket                         |
| Expert Prediction       | HIGH_RISK         | Decision Tree output                                    |
| Clinical Class          | NORMAL_MUC<br>OSA | Normal mucosal pattern — no polyp morphology identified |
| Detection Confidence    | 80.9%             | YOLO / RT-DETR / track confidence                       |

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #2 — LIFTED\_POLYP [LOW\_RISK] (Tracked across 152 frames)



| Feature          | Value                               | Interpretation                                       |
|------------------|-------------------------------------|------------------------------------------------------|
| Frame Index      | 2010                                | Frame 2010 of 3006                                   |
| Detection Method | Detector Consensus (YOLO / RT-DETR) | Model Agreement (MedSAM2 unavailable for this video) |
|                  |                                     |                                                      |
| POLYP TYPE       | LIFTED_POLYP                        | Type confidence: 66.8%                               |
|                  |                                     |                                                      |

|                         |                   |                                                          |
|-------------------------|-------------------|----------------------------------------------------------|
| Redness Score           | 0.058             | Color-based vascularization                              |
| Relative Radius         | 22.3%             | Polyp size as % of frame diagonal                        |
| Texture Score           | 0.144             | Surface morphology                                       |
| Vessel Visibility       | 0.000             | Vascularization indicator                                |
| Risk Tier               | LOW RISK          | Validated by ESGE 2017 / NICE classification             |
|                         |                   |                                                          |
| Clinical Classification | HIGH_RISK         | Final symbolic reasoning bucket                          |
| Expert Prediction       | HIGH_RISK         | Decision Tree output                                     |
| Clinical Class          | ADENOMATOUS_POLYP | Polypoid lesion — discrete mucosal protrusion identified |
| Detection Confidence    | 85.7%             | YOLO / RT-DETR / track confidence                        |

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #3 — LIFTED\_POLYP [LOW\_RISK] (Tracked across 180 frames)



| Feature          | Value                               | Interpretation                                       |
|------------------|-------------------------------------|------------------------------------------------------|
| Frame Index      | 2968                                | Frame 2968 of 3006                                   |
| Detection Method | Detector Consensus (YOLO / RT-DETR) | Model Agreement (MedSAM2 unavailable for this video) |
|                  |                                     |                                                      |
| POLYP TYPE       | LIFTED_POLYP                        | Type confidence: 58.9%                               |

|                         |                   |                                                          |
|-------------------------|-------------------|----------------------------------------------------------|
|                         |                   |                                                          |
| Redness Score           | 0.005             | Color-based vascularization                              |
| Relative Radius         | 14.8%             | Polyp size as % of frame diagonal                        |
| Texture Score           | 0.158             | Surface morphology                                       |
| Vessel Visibility       | 0.000             | Vascularization indicator                                |
| Risk Tier               | LOW RISK          | Validated by ESGE 2017 / NICE classification             |
|                         |                   |                                                          |
| Clinical Classification | HIGH_RISK         | Final symbolic reasoning bucket                          |
| Expert Prediction       | HIGH_RISK         | Decision Tree output                                     |
| Clinical Class          | ADENOMATOUS_POLYP | Polypoid lesion — discrete mucosal protrusion identified |
| Detection Confidence    | 83.5%             | YOLO / RT-DETR / track confidence                        |

**Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.**

*Low-confidence LOW RISK detection - Follow standard protocol*

**POLYP #4 — LIFTED\_POLYP [LOW\_RISK] (Tracked across 26 frames)**



| Feature                 | Value                                  | Interpretation                                           |
|-------------------------|----------------------------------------|----------------------------------------------------------|
| Frame Index             | 1867                                   | Frame 1867 of 3006                                       |
| Detection Method        | Detector Consensus (MedSAM2 / RT-DETR) | MedSAM2 Unavailable (MedSAM2 unavailable for this video) |
|                         |                                        |                                                          |
| POLYP TYPE              | LIFTED_POLYP                           | Type confidence: 67.0%                                   |
|                         |                                        |                                                          |
| Redness Score           | 0.073                                  | Color-based vascularization                              |
| Relative Radius         | 12.9%                                  | Polyp size as % of frame diagonal                        |
| Texture Score           | 0.139                                  | Surface morphology                                       |
| Vessel Visibility       | 0.000                                  | Vascularization indicator                                |
| Risk Tier               | LOW RISK                               | Validated by ESGE 2017 / NICE classification             |
|                         |                                        |                                                          |
| Clinical Classification | HIGH_RISK                              | Final symbolic reasoning bucket                          |
| Expert Prediction       | HIGH_RISK                              | Decision Tree output                                     |
| Clinical Class          | ADENOMATOUS_POLYP                      | Polypoid lesion — discrete mucosal protrusion identified |
| Detection Confidence    | 83.3%                                  | YOLO / RT-DETR / track confidence                        |

**Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.**

*Low-confidence LOW RISK detection - Follow standard protocol*



# CLINICAL IMPRESSION

## Summary:

A total of 4 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

## Risk Distribution:

- HIGH RISK polyps: 0 (0.0%)
- MEDIUM RISK polyps: 0 (0.0%)
- LOW RISK polyps: 4 (100.0%)

## Recommendation:

All detected polyps are classified as LOW RISK. Standard surveillance protocol is recommended.

## Technical Details:

- Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
- Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
- Classification: Cluster-specific Decision Tree experts
- Confidence: Based on deep learning + medical heuristics

**Disclaimer:** This report is generated by an AI-assisted research system for academic and research purposes only. All clinical recommendations are derived from published ESGE 2017, NICE CG118/CG131, and BSG 2019 colonoscopy guidelines. This output does not constitute medical advice and must not be used as the sole basis for clinical decisions. All findings must be reviewed and confirmed by a qualified gastroenterologist or endoscopist.